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Research on Remote Sensing Recognition of Forest Fire Smoke Based on Machine Learning

Kaihua Wang
College of Geoexploration Science and Technology
Jilin University
Changchun, China
976966022@qq.com

Jun Pan
College of Geoexploration Science and Technology
Jilin University
Changchun, China
panj@jlu.edu.cn

Lijun Jiang *
College of Geoexploration Science and Technology
Jilin University
Changchun, China
Corresponding author: jlijun@jlu.edu.cn

Yehan Sun
College of Geoexploration Science and Technology
Jilin University
Changchun, China
yhsun@mails.jlu.edu.cn

Kaisi Wang
College of Geoexploration Science and Technology
Jilin University
Changchun, China
1285522406@qq.com

Yu Cao
Natural Resources Department, Heilongjiang Provincial Bureau of Surveying and Mapping Geographic Information
Heilongjiang Fundamental Geographic Information Center
Harbin, China
2471320169@qq.com

***Abstract*—To investigate the differences in spectral features between smoke and other typical features, and to achieve remote sensing recognition of forest fire smoke. This paper takes muli county, liangshan, Sichuan province as the study area, uses Sentinel 2 multispectral images as the data source, extracts smoke, clouds, water bodies, vegetation, and bare ground image elements samples, builds forest fire smoke recognition models based on machine learning algorithms such as decision tree algorithm, support vector machine algorithm, and neural network algorithm, extracts the sum parameter of the reflection peak difference(SPRPD) and the product parameter of the reflection peak difference(PPRPD) to improve the models. The experimental results show that the recognition accuracies of the three models are 97.3%, 99.2%, and 99.4%, respectively, and the neural network model has the highest accuracy. The improved smoke recognition model reduces the probability of misclassification of thin clouds into smoke, and the model accuracies are improved to 99.0%,99.5%and 99.9%, respectively. The study shows that the combined feature parameters of the reflection peak band can effectively distinguish thin clouds and smoke, improve the accuracy of smoke recognition, and the smoke recognition model based on the neural network algorithm can effectively identify forest fire smoke.**

Keywords-smoke; Sentinel-2; Neural network algorithm; Characteristic parameters;

I. INTRODUCTION

A forest fire is a fire that loses human control and spreads in the forest land, which is a very destructive natural disaster that not only causes trees to burn, but plants and animals to lose their homes to the fire, and affects the forest's ecological environment, but also poses a threat to human personal safety, social and economic development [1]. Forest fires release a large amount of smoke due to the burning of vegetation, and smoke is an important signal of fires because of its rapid diffusion and widespread [2], and the detection and identification of fire smoke can detect fires in time.

Scholars at home and abroad have studied forest fire smoke identification, based on the feature differences between smoke and other feature types for smoke detection and identification, Baum et al. used the threshold method to identify NOAA/AVHRR remote sensing images containing feature types such as clouds, fire points, smoke and snow [3]; Xie et al. used MODIS data as the basis and used the multi-channel threshold method to achieve the forest fire smoke detection [4]; Li Xiaolian et al. proposed a forest fire smoke identification algorithm based on smoke, cloud, and subsurface spectral analysis based on MODIS data, combined with multi-channel thresholding method and BP network classifier to carry out effective smoke monitoring [5]. Traditional recognition methods use manual visual interpretation, which is labor-intensive and cannot automatically and quickly process large amounts of remotely sensed data; recognition methods based on thresholding [6] need to determine different optimal thresholds according to specific study areas and seasons, which is not highly applicable; Recognition methods based on machine learning [7] can fully learn to use the differences in spectral features between smoke and other typical features to automatically smoke recognition, with high recognition accuracy and speed. In the smoke identification study, the

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spectral features of smoke and thin clouds are similar, and how to accurately distinguish them deserves further study.

Therefore, this study establishes a forest fire smoke recognition model based on a machine learning method to detect and identify forest fire smoke. Using muli county of liangshan mountain as the study area, we extract smoke, cloud, water, vegetation, and bare ground image samples using Sentinel-2 multispectral images as the data source, quantitatively analyze the spectral feature differences between fire smoke and other typical features using analysis of variance method, establish smoke recognition model based on decision tree algorithm, support vector machine algorithm, and neural network algorithm, analyze the recognition results, extract waveband combination feature parameters to improve the model to monitor and identify the forest fire smoke, to provide a reference for timely rescue of forest fires.

II. Materials and Methods

A. Research Area

Muli Tibetan Autonomous County of Sichuan Province belongs to Liangshan Yi Autonomous Prefecture of Sichuan Province, located in southwest Sichuan province, the southeastern edge of the Qinghai-Tibet Plateau, the end of the Hengduan Mountains, with an average altitude of 3,100 meters.

B. Data sources and data processing

At 18:00 on March 30, 2019, a forest fire broke out in Lier Village, Yalongjiang Town, Muli County, Sichuan Province, causing huge casualties and a total overfire area of about 20 hectares. The Sentinel 2A satellite transited in Muli County, Liangshan Prefecture, Sichuan Province, on March 31, 2019, the time phase was similar to the time of the forest fire, and the location of the forest fire on the remote sensing image was unobstructed by clouds, with obvious smoke, so the vegetation and smoke distribution area that was still burning the day after the fire was selected as the study area. The radiometric calibration, atmospheric correction, cropping, resampling, and other operations were performed on the images to complete the pre-processing of the image data. The remote sensing images are shown in Figure 1.

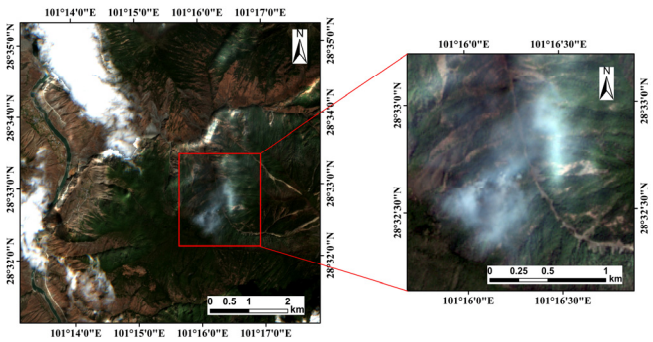

Figure 1 Remote sensing image of 3.31 at the time of forest fire in Liangshan

C. Research Methods

1) Sampleset Extraction： To identify forest fire smoke, based on the visual discrimination method, five image element sample datasets of typical feature categories were extracted from remote sensing images, including smoke, clouds, vegetation, water bodies, and bare ground.

2) Spectral characterization： To study the separability of smoke from other feature classes, we compare and analyze the spectral characteristics of smoke with those of clouds, water bodies, vegetation, bare ground, and other features, and use the analysis of variance method to calculate the separability measures of smoke image samples from other typical feature samples in each band and to quantify the differences in spectral characteristics between smoke image elements and other typical features.

Analysis of variance is used to analyze the importance of influencing factors based on the variability characteristics of the data [8]. For a certain band, the difference in spectral data between different feature categories is the intergroup variance, and the difference in spectral data of the same feature category is the intra-group variance, if the inter-group variance is significantly larger than the intra-group variance, it means that the difference of two feature categories under that band is obvious and can be well separated and identified, and the band has good separability for two features, and the separability measure S is introduced to indicate the separability between features.

3) Machine learning classification methods A decision tree algorithm is a typical classification algorithm that approximates the value of a discrete function and classifies the data by a series of rules, presenting the classification process in the form of a tree.

Support vector machine algorithm is a supervised recognition method used for regression analysis and classification, based on statistical training, mapping sample data to a high-dimensional space and then finding an optimal hyperplane that minimizes the classification error and maximizes the interval between categories.

A neural network algorithm is a mathematical model that simulates the structure and function of a biological neural network, a nonlinear complex network system consisting of a large number of simple neurons connected.

Based on machine learning algorithms such as decision trees, support vector machines, and neural networks, a classification and recognition model for forest fire smoke is established.

III. Results and Analysis

A. Results of spectral characterization

In this study, image element samples of smoke, clouds, vegetation, water bodies, and bare ground were extracted, and the average spectral reflectance curves of the five feature types are shown in Figure 2. It can be seen from the figure that the spectral reflectance of smoke shows a trend of decreasing, then increasing, and then decreasing with the increase of wavelength, with a reflection valley at 664.5 nm in the red band and a

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reflection peak at 864.8 nm in the near-infrared band; the spectral reflectance of clouds is higher in all bands, higher than the other four types of features, with a larger reflection peak at 945 nm; and the reflectance of water is higher in all bands. The spectral reflectance of vegetation and the bare ground tends to increase and then decrease with the increase of wavelength, and the spectral reflectance of vegetation reaches the highest in the NIR band at 864.8 nm, while the spectral reflectance of bare ground reaches the highest in the short-wave IR band at 1613.7 nm.

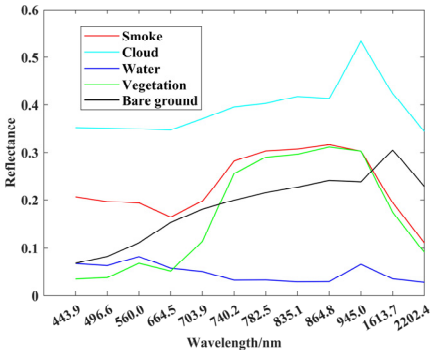

Figure 2 Average spectral reflection curves of five object types

The spectral reflectance curves of smoke and clouds, water bodies, vegetation, and bare ground are shown in Figure 3, from which it can be seen that the spectral reflectance of clouds is generally higher than that of smoke, but the spectral reflectance of cloud image samples located at the edge of clouds, i.e., when the clouds are relatively thin, is not much different from that of smoke. For all bands, the spectral reflectance of smoke is higher than that of water, and the spectral characteristics of the two are different; for smoke and vegetation, in the visible band, the spectral reflectance of smoke is higher than that of vegetation, and as the wavelength increases, the sunlight penetrates more into the smoke. For smoke and vegetation, the spectral reflectance of smoke is higher than that of vegetation in the visible wavelengths, and the spectral reflectance of smoke is higher than that of vegetation in the short-wave infrared wavelengths. The spectral reflectance of smoke was higher than that of bare ground, and the spectral reflectance of smoke showed a decreasing trend while that of the bare ground showed an increasing trend, and the difference between the two spectral features became smaller as the wavelength increased, but in the short-wave infrared band of 1613.7 nm, there was an emission peak in the bare ground image sample, but there was no such feature in the smoke image sample.

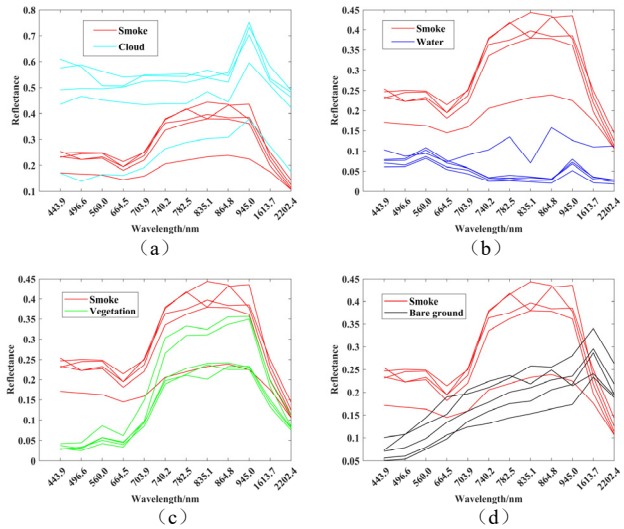

Figure 3 Spectral reflectance of smoke and other typical features

B. *Separability metric analysis*

The separability measures of smoke and cloud, water, vegetation, and bare ground samples in each band were calculated, and the results are shown in Figure 4, which shows that the separability measures of smoke and water are high in all bands, indicating that smoke and water can be distinguished in all bands. 664.5 nm, 703.9 nm and other visible bands, and 945 nm, 1613.7 nm, 2202.4 nm short-wave infrared bands are higher, and can distinguish the two better; while for the distinction between smoke and vegetation, the separability measures are higher in 443.9 nm, 496.6 nm, 560.0 nm, 664.5 nm bands, while the other bands Due to the strong penetration ability of smoke, most of the radiant energy received by the satellite sensor comes from the reflected energy of the background features under the smoke mat, so the spectral characteristics of the two are similar and the separability is poor; for smoke and bare ground, the separability metric values are better in the wavelengths of 443.9 nm, 496.6 nm, 560.0 nm bands, indicating that these bands can distinguish the two better, although the wavelengths in However, due to the penetration of sunlight into the smoke, the spectral characteristics of the smoke in these bands are actually the spectral characteristics of the background features in the subsurface, which in this study are the spectral characteristics of the vegetation, so the difference between the spectral characteristics of the smoke and the bare ground in the near-infrared band and the short-wave infrared band are actually the spectral characteristics of the vegetation and the bare ground. In this study, the difference in spectral characteristics between smog and bare ground is the difference in spectral characteristics between vegetation and bare ground.

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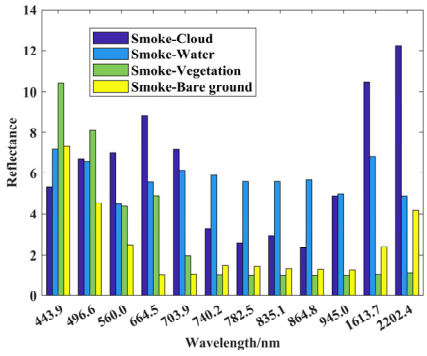

Figure 4 Separability metrics for smoke and other typical features

C. Model Construction

Based on machine learning algorithms including decision trees, support vector machines, and neural networks to build a smoke recognition model, the spectral reflectance of the samples is used as input to train the smoke recognition model, and at the same time, to prevent the model from overfitting, a 5-fold cross-validation method is used to divide the data set into 5 parts, using the other 4 parts in turn as training and the remaining one as testing, and finally taking the average test error as the final error.

The model accuracy of the decision tree is the lowest at 97.3%, the recognition accuracy of the support vector machine model is 99.2%, and the recognition accuracy of the neural network model is the highest at 99.4%.

The confusion matrix is used to describe the situation of each feature being correctly recognized and misclassified into other feature categories, and to summarize the prediction results of the classification model. The confusion matrix of the model training is shown in TABLE I， TABLE II, and TABLE III, the horizontal coordinates represent the predicted feature classes and the vertical coordinates represent the actual feature classes. From the table, we can see that in the decision tree model, the misclassification error of smoke is 4.33% and the omission error is 2.83%; for the support vector machine model, the misclassification error of smoke is 3.06% and the omission error is 0.31%; for the neural network model, the misclassification error of smoke is 1.86% and the omission error is 0.63%. In the three smoke recognition models, some samples were misclassified or omitted.

TABLE I. DECISION TREE MACHINE MODEL CONFUSION MATRIX

Actual category	Prediction Category				
	Smoke	Cloud	Water	Vegetation	Bare ground
Smoke	309	6	0	3	0
Cloud	12	360	0	1	0
Water	2	0	283	1	0
Vegetation	0	0	0	282	12
bare ground	0	0	0	5	303

TABLE II. SUPPORT VECTOR MACHINE CONFUSION MATRIX

Actual category	Prediction Category				
	Smoke	Cloud	Water	Vegetation	Bare ground
Smoke	317	1	0	0	0
Cloud	9	363	0	0	1
Water	1	0	285	0	0
Vegetation	0	0	0	294	0
bare ground	0	0	0	0	308

TABLE III. NEURAL NETWORK MODEL CONFUSION MATRIX

Actual category	Prediction Category				
	Smoke	Cloud	Water	Vegetation	Bare ground
Smoke	316	2	0	0	0
Cloud	5	368	0	0	0
Water	1	0	285	0	0
Vegetation	0	0	0	293	1
bare ground	0	0	1	0	307

D. Model Improvement

Through the analysis of the confusion matrix results of the model, the type of misclassification and omission is mainly cloud misclassification into smoke, and the spectral curve of the cloud sample that is misclassified into smoke is shown in Figure 5, from which it can be seen that the misclassified cloud sample is the thin position of the cloud layer, and its spectral reflectance is lower than the average spectral reflectance of the cloud sample and is at the same level with the spectral reflectance of smoke.

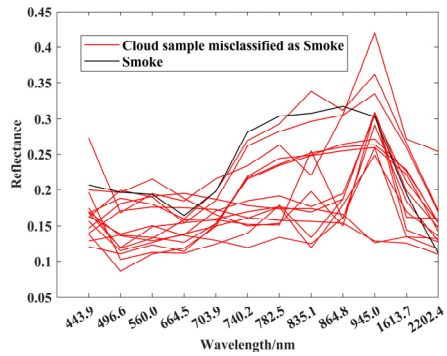

Figure 5 Spectral reflection curve of cloud sample misclassified as smoke

After the spectral feature analysis, the cloud image element sample has an obvious spectral reflection peak at the 945 nm band, and the reflection peak feature can be extracted to distinguish the smoke and thin clouds.

To quantify the reflection peak feature, the sum and product parameters of the difference between the reflectance of the band at the peak location and the reflectance of the two adjacent bands are introduced, that is, the sum parameter of the reflection peak difference(SPRPD) and the product parameter of the reflection peak difference(PPRPD), when both SPRPD and PPRPD are greater than 0, there is a reflection peak in this band range, $R_{864.8}$, $R_{945.0}$ and 、 $R_{1613.7}$ refers to the spectral reflectance of the band with the central wavelength of 864.8nm, 945.0nm, and 1613.7nm respectively.

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$$SPRPD = (R_{945.0} - R_{864.8}) + (R_{945.0} - R_{1613.7}) \quad (1)$$

$$PPRPD = (R_{945.0} - R_{864.8}) \times (R_{945.0} - R_{1613.7}) \quad (2)$$

The smoke recognition model was trained by adding SPRPD and PPRPD to the smoke recognition model, together with spectral reflectance as the model input. After training, the recognition accuracy of the improved decision tree recognition model is 99.0%, the recognition accuracy of the improved support vector machine model is 99.5%, the recognition accuracy of the improved neural network model is 99.9%, and the recognition accuracy of all three models has been improved.

The confusion matrix of model training is shown in TABLE IV、TABLE V and TABLE VI. From the table, it can be seen that in the decision tree improved model, the misclassification error of smoke is 1.89%, and the omission error is 2.20%; For the support vector machine model, the misclassification error of smoke is 2.45%, the omission error is 0; For the neural network model, the misclassification error is 0, the omission error is 0.31%.

TABLE IV. IMPROVED DECISION TREE MACHINE CONFUSION MATRIX

Actual category	Prediction Category				
	Smoke	Cloud	Water	Vegetation	Bare ground
Smoke	311	5	0	2	0
Cloud	6	366	0	1	0
Water	0	2	284	0	0
Vegetation	0	0	0	294	0
bare ground	0	0	0	0	308

TABLE V. IMPROVED SUPPORT VECTOR MACHINE CONFUSION MATRIX

Actual category	Prediction Category				
	Smoke	Cloud	Water	Vegetation	Bare ground
Smoke	318	0	0	0	0
Cloud	7	366	0	0	0
Water	1	0	285	0	0
Vegetation	0	0	0	294	0
bare ground	0	0	0	0	308

TABLE VI. IMPROVED NEURAL NETWORK CONFUSION MATRIX

Actual category	Prediction Category				
	Smoke	Cloud	Water	Vegetation	Bare ground
Smoke	317	1	0	0	0
Cloud	0	373	0	0	0
Water	0	1	285	0	0
Vegetation	0	0	0	294	0
bare ground	0	0	0	0	308

By adding SPRPD and PPRPD, which can characterize thin clouds, the model's accuracy is improved, and the probability of misclassification and omission is reduced, which makes the smoke recognition more accurate.

The smoke recognition model based on the neural network algorithm is used to detect and identify the smoke of forest fires in Liangshan Mountain, and the recognition results are shown in Figure 6, from which it can be seen that the smoke location and contour information are accurately identified.

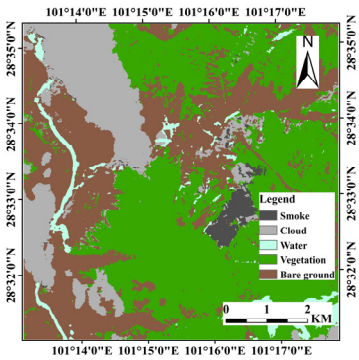

Figure 6 Smoke identification results of forest fires in Liangshan

IV. CONCLUSIONS

This paper used the Sentinel-2 multispectral satellite as the data source and Muli County, Liangshan Prefecture, Sichuan Province as the research area, and extracted typical pixel samples of smoke, clouds, water, vegetation, and bare land. In this paper. By analyzing the spectral feature differences between samples, based on machine learning algorithms such as decision trees, support vector machines, and neural networks, a forest fire smoke recognition model is established, and the band combination parameters are extracted to improve the model.

(1) The spectral reflectance of most clouds is greater than the smoke, but when the clouds are thin, the spectral reflectance difference between clouds and the smoke is not great, and there is a spectral reflectance peak in the 945nm wavelength range of clouds, which can be used for the identification of the smoke and thin clouds; the spectral features of the smoke and water bodies and bare ground have great differences, but with the wavelength increases, the spectral features of the smoke in the near-infrared and short-wave infrared bands are similar to the spectral features of the background features on the subsurfac.

(2) Based on the machine learning algorithm of the decision tree, support vector machine, and neural network, smoke recognition models were developed for forest fires, and the models were trained using spectral bands as input. To improve the accuracy of smoke recognition, we added SPRPD and PPRPD to improve the model, and through the accuracy comparison analysis, the improved smoke recognition model reduced the probability of thin clouds being misclassified as smoke, and the accuracy of the model increased to 99.0%, 99.5%, and 99.9%, which made the smoke recognition more accurate. The smoke recognition model based on the neural network algorithm detects and identifies the smoke of forest fires in Liangshan, and the smoke location and contour information are accurately identified. The study shows that the combined feature parameters of the reflection peak band can effectively distinguish thin clouds from smoke and improve the accuracy of smoke identification; the smoke identification model based on a neural network algorithm can effectively identify smoke and provide a reference for fire rescue.

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